

Section 5.1 Worksheet

1. Find the norm and a unit vector in the direction of $(6, -3, 6)$.
2. Find the distance between $(1, 4, -2)$ and $(5, -2, 1)$.
3. Compute the dot product and angle between $(1, 2, 2)$ and $(2, 0, -1)$.
4. Determine whether $(2, -1, 4)$ and $(1, 6, 1)$ are orthogonal.
5. Verify the Cauchy–Schwarz inequality for $(2, 1)$ and $(-1, 3)$.
6. Write the dot product of (a, b, c) and (x, y, z) as a matrix product.

Section 5.2 Worksheet

1. For $\langle u, v \rangle = 3u_1v_1 + 2u_2v_2$, find $\langle (2, -1), (4, 3) \rangle$ and the induced norm of $(2, -1)$.
2. Explain why $u_1v_1 + u_2v_2 - u_3v_3$ is not an inner product on $\mathbb{R}^3$.
3. Using the ordinary coefficient inner product, find $\langle 1 + x + 2x^2, 2 - 3x + x^2 \rangle$.
4. On $C[0, 1]$, compute $\langle 1, x^2 \rangle = \int_0^1 x^2 dx$.
5. Find the projection of $(5, 1, 4)$ onto $(1, 0, 2)$.
6. Verify that the residual in Problem 5 is orthogonal to the projection direction.

Section 5.3 Worksheet

1. Determine whether $\{(1, 1, 0), (1, -1, 0), (0, 0, 2)\}$ is orthogonal. Is it orthonormal?
2. Normalize the set in Problem 1.
3. Find the coordinates of $(4, 2, 3)$ relative to the normalized basis.
4. Apply Gram–Schmidt to $(1, 0)$ and $(1, 2)$.
5. Apply Gram–Schmidt to $(1, 1, 0)$ and $(1, 0, 1)$.
6. Find an orthonormal basis for $\text{span}\{(1, 1, 1), (1, -1, 0)\}$.

Section 5.4 Worksheet

1. Find a basis for $\text{span}\{(1, 1, 0), (0, 1, 1)\}^\perp$.
2. Project $(4, 5, 6)$ onto the line spanned by $(1, 1, 1)$.
3. Project $(3, 1, 4)$ onto the plane spanned by $(1, 1, 0)$ and $(1, -1, 2)$.
4. Find rank, row-space basis, column-space basis, and nullspace basis for $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$.
5. Find the least-squares line for $(0, 1), (1, 2), (2, 2), (3, 5)$.
6. Verify that the residual from Problem 5 is orthogonal to both design-matrix columns.

Section 5.5 Worksheet

1. Find $(2, 1, -1) \times (1, -2, 3)$.
2. Verify that the answer to Problem 1 is orthogonal to both original vectors.
3. Find the parallelogram and triangle areas determined by the vectors in Problem 1.
4. Find a unit normal to the plane spanned by $(1, 0, 1)$ and $(0, 2, 1)$.
5. Find the best constant approximation to $f(x) = x$ on $[0, 2]$.
6. Write the second-order Fourier approximation of $f(x) = x - \pi$ on $[0, 2\pi]$.